



Integrating Environmental Sustainability and Operational Safety in Small-Scale Purse Seine Fisheries: A CCRF-Aligned Risk Assessment Framework from East Java, Indonesia

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Abstract: In East Java, small-scale purse seine fisheries play a critical role in preserving food security, local economies, and coastal cultural systems; however, there are challenges of sustainability due to the dual pressures of ecological responsibility and risks of operational safety. This study evaluated purse seine operations at Tambakrejo Fishing Port, a key landing site in East Java, by integrating Food and Agriculture Organization Code of Conduct for Responsible Fisheries (FAO-CCRF) sustainability indicators with a Frequency–Severity Index (FI–SI) occupational risk assessment. The CCRF indicators were anchored in Articles 6–8 (general principles, fisheries management, and fishing operations) to examine ecological performance and exposure to hazards across distinct operational phases. Data were collected through direct observation, structured fisher interviews, port documentation, and catch monitoring (9 vessels over 15 sampling days). FI–SI scores were assigned by a standardized rubric triangulated with evidence from interviews and port records. Results indicated a highly selective catch composition with minimal bycatch (~0.4%), though species-specific vulnerabilities persisted due to sub-length at first maturity (L_m) retention, particularly in *Euthynnus affinis*. Risk evaluation showed that the highest hazard exposure occurred during labor-intensive and time-pressure phases such as setting, pursing, and hauling, driven by rope handling, wet-deck dynamics, and repetitive manual tasks. The proposed dual-matrix approach differed from certification-oriented indicator sets (e.g., MSC-type schemes), Ecosystem Approach to Fisheries Management (EAFM) scorecards, and standalone occupational health and safety matrices by linking phase-level ecological signals with task-level safety risk to identify high-risk–low-compliance nodes and prioritise feasible controls. The integration of sustainability and risk indicators suggests that compliant and selective practices could reduce both ecological pressure and hazard exposure, hence upholding the concept that sustainability and safety are mutually reinforcing outcomes. The framework offers practical guidance for adaptive co-management by emphasizing low-cost improvements, training, and procedural discipline, while acknowledging that cross-sectional sampling, seasonal variability of sea states, and local implementation capacity could influence risk profiles and its feasibility.

Keywords: Small-scale fisheries; Purse seine; Food and Agriculture Organization Code of Conduct for Responsible Fisheries; Risk assessment; Sustainability management; Decision support

1. Introduction

Small-scale fisheries represent one of the most vital sectors sustaining coastal livelihoods, particularly across developing maritime nations where access to marine resources underpins both food security and economic

resilience. In Indonesia, this sector plays an indispensable role, with approximately 2.7 million people depending directly on fishing for their primary income; of whom over 80% operate within small-scale fisheries (Alghozali et al., 2025). These fisheries not only secure access to affordable and nutrient-rich seafood but also stabilize household economies in regions with limited alternative employment (Andreoli et al., 2025; Gibson et al., 2020). In many coastal provinces such as East Java, small-scale purse seine fleets contribute substantially to the supply of pelagic fish, particularly sardines, mackerel, and small tuna species, hence supporting local markets and community-based processing industries. Beyond their economic significance, these fisheries serve essential social and cultural functions, to strengthen local cohesion through shared labor practices and intergenerational knowledge transfer (Andreoli et al., 2025). Yet the same operational conditions that enable high catch efficiency could simultaneously shape ecological outcomes (e.g., selectivity) and human outcomes (e.g., injury risk), contributing to sustainability and safety co-produced properties of fishing operations rather than separate policy tracks. Given these contributions, the sustainability of small-scale purse seine fisheries is not only a sectoral concern but also a critical dimension of coastal socio-economic stability and long-term marine resource stewardship.

The socio-economic resilience of small-scale fisheries is further reinforced by livelihood diversification and community-level adaptability. Fishers who engage in multiple fishing methods or seasonal activities could be better buffers against market volatility and environmental uncertainty. In comparative studies, diversification has been shown to mitigate income instability caused by climatic variations or resource fluctuations, as observed in the Golfo de Ulloa and similar artisanal systems (Mason et al., 2021). Within Indonesia, collective management and co-governance structures, wherein communities share decision-making authority with local governments, have proven effective in enhancing fisheries performance, resource stewardship, and adaptive capacity (Hidayat et al., 2022; Mustika et al., 2020). Consequently, sustainability in this sector must be understood through a triple-bottom-line lens, i.e., environmental, social, and economic, to align with contemporary global frameworks such as the Sustainability Development Goals (SDGs), blue economy principles, and climate-resilient coastal governance frameworks. However, widely used assessment approaches often remain siloed: certification-oriented indicator sets (e.g., MSC-type schemes) and Ecosystem Approach to Fisheries Management (EAFM) scorecards emphasize ecological and governance dimensions, while occupational health and safety (OHS) matrices focus on hazard control without explicitly embedding responsible-fisheries policy indicators. This separation limits decision support at the operational level, in which managers and fishers must prioritize concrete phase-specific actions under resource constraints.

Sustainability of small-scale fisheries in Indonesia cannot be assessed solely through biological indicators, but should also consider socio-cultural resilience and community-based wisdom, particularly those embedded in traditional norms and knowledge systems that contribute to ecosystem protection and sustainable harvesting practices (Parsaulian et al., 2023). This understanding aligns with recent sustainability literature which emphasized that adaptive and culturally-rooted models played a critical role in maintaining resource continuity in Indonesian fisheries. Despite their essential contributions, small-scale purse seine fisheries are characterized by high levels of occupational risk and operational vulnerability. Fishing remains one of the most hazardous maritime occupations globally, with a high incidence of physical injuries, vessel accidents, and fatalities (Roy et al., 2024). Common hazards include musculoskeletal strain from repetitive hauling, entanglement in gear, and slips or falls on wet decks, exacerbated by prolonged working hours, sea-state variability (e.g., monsoon vs. calmer periods), and limited onboard safety infrastructure (Seto et al., 2023). Accordingly, this study introduced an operational-phase framework that mapped Food and Agriculture Organization Code of Conduct for Responsible Fisheries (FAO-CCRF) domains to discrete purse-seine phases and overlaid Frequency–Severity Index (FI–SI) scoring at the task level. This produced a dual-matrix output that supported prioritisation of no-regret interventions (low-cost procedural controls and training) while explicitly acknowledging implementation constraints (enforcement capacity, informal labour arrangements, and affordability) that shaped feasibility in the context of a developing country. The integration logic is summarised in Figure 1, illustrating the CCRF-to-phase mapping, the FI–SI scoring point, and how combined outputs translate into management actions.

Psychosocial stressors also play a critical role, as fishers often contend with income uncertainty and competition from industrial fleets, which intensify economic insecurity and mental fatigue (Eurich et al., 2023). In Indonesia, limited formalization of occupational safety arrangements and constrained access to health insurance could amplify these risks, leaving fishing households vulnerable to both immediate injury and long-term financial distress (Mir et al., 2023; Muslim et al., 2023). These occupational threats not only endanger fishers but also have indirect consequences on long-term sustainability by reducing labor capacity, increasing economic vulnerability, and accelerating social instability within fishing communities. To make this linkage actionable, the present study operationalized the safety–sustainability intersection at the level of fishing operations, rather than treating “safety” as a separate social issue. This intersection is still under-represented in mainstream fisheries assessment tools although evidence suggests that unsafe operational practices could trigger social and ecological feedback loops that undermine stock productivity and management goals. By integrating CCRF-aligned indicators with phase-resolved FI–SI scoring, the framework provides a decision-support pathway to identify tasks in which improving procedural discipline and selectivity could yield simultaneous ecological and human-safety benefits.

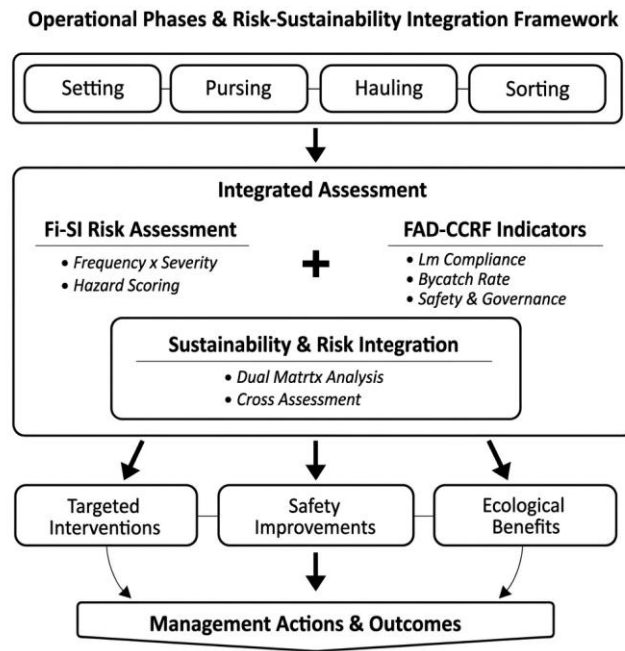


Figure 1. Conceptual workflow of the integrated Food and Agriculture Organization Code of Conduct for Responsible Fisheries (FAO-CCRF) sustainability–Frequency–Severity Index (FI–SI) occupational risk assessment for small-scale purse seine fisheries

Technological and environmental factors further influence the sustainability of purse seine operation. These vessels rely heavily on manual labor with limited mechanization, thus increasing exposure to hazards during net deployment, hauling, and sorting. Monsoon-driven oceanographic changes also affect operational risks and catch outcomes (Roy et al., 2024). Since the present dataset was cross-sectional, seasonal variability was treated explicitly as a representativeness constraint rather than assumed away. Integrating such environmental variability into sustainability and risk assessment frameworks was crucial, as fluctuations of climate-driven sea state could exacerbate both ecological pressure and occupational hazards, thus placing small-scale fleets at the frontline of compound sustainability challenges (Halder et al., 2024). The framework was designed accordingly so that FI–SI scores could be re-applied across contrasting sea-state regimes (e.g., monsoon vs. calmer periods) to test whether risk rankings at phase level and selectivity outcomes remained stable or shift materially.

Therefore, this study seeks to strengthen the conceptual and practical link between environmental sustainability and occupational safety by developing a CCRF-aligned, operational-phase dual-assessment approach for small-scale purse seine fisheries. To address the identified gaps in the decision support at operational level, this research advanced three integrated objectives. Specifically, the study (i) mapped CCRF domains and selected indicators to discrete purse-seine operational phases; (ii) applied FI–SI scoring at the task level in each phase; and (iii) generated a dual-matrix output that prioritised feasible interventions under the constraints in real world. This integrated perspective aims to inform sustainable fisheries governance, improve risk decision making in operation, and contribute to resilient coastal community development, including contexts where enforcement capacity, informal labour arrangements, and affordability shape the feasibility of implementation.

2. Methods

This study employed an integrated analytical framework combining the FAO-CCRF sustainability indicators with the FI–SI risk assessment approach to evaluate environmental sustainability and operational safety in small-scale purse seine fisheries across East Java. The methodology was designed to provide a holistic understanding of how ecological performance, vessel operations, and occupational hazards interact within artisanal fishing systems. As summarised in Figure 1, the workflow proceeds from the acquisition of field data (observations, interviews, port records, and catch monitoring) to two parallel assessments: (i) CCRF-aligned sustainability indicators (e.g., bycatch ratio, compliance of length at first maturity (Lm), habitat interaction, safety capacity, and governance compliance) and (ii) phase-based FI–SI hazard scoring across purse-seine operational phases (setting, pursing, hauling, sorting, and landing). The two outputs are then overlaid at the operational-phase level to generate a dual-matrix decision-support result that links ecological compliance with safety exposure and translates it into prioritised management actions (procedural controls, training, and low-cost technical improvements) for adaptive co-management. This design supports transparent replication by making the phase mapping, scoring locus, and

integration step explicit.

The study was conducted at Tambakrejo fishing ports (Figure 2), with a focus on sites where small-scale purse seine fleets dominated local fisheries. These locations representing critical hubs of pelagic fish landings were selected for their economic importance and representative operational conditions. The fieldwork encompassed observations of vessel activities, interviews with fishers, and measurements of fleet characteristics, including vessel size, engine power, crew composition, and operational routines. Environmental variables such as sea state, patterns of weather, and fishing grounds were also documented to establish correlations between operational behavior and exposure to environmental risks. To address temporal representativeness, the sampling period was reported explicitly, and seasonal effects (monsoon vs. calmer periods) were treated as a limitation; FI–SI scoring was therefore interpreted as a snapshot for the observed window rather than a seasonal average. This contextualized approach enabled the results to reflect both biophysical realities and human-centered sustainability dynamics that characterized small-scale fisheries systems.

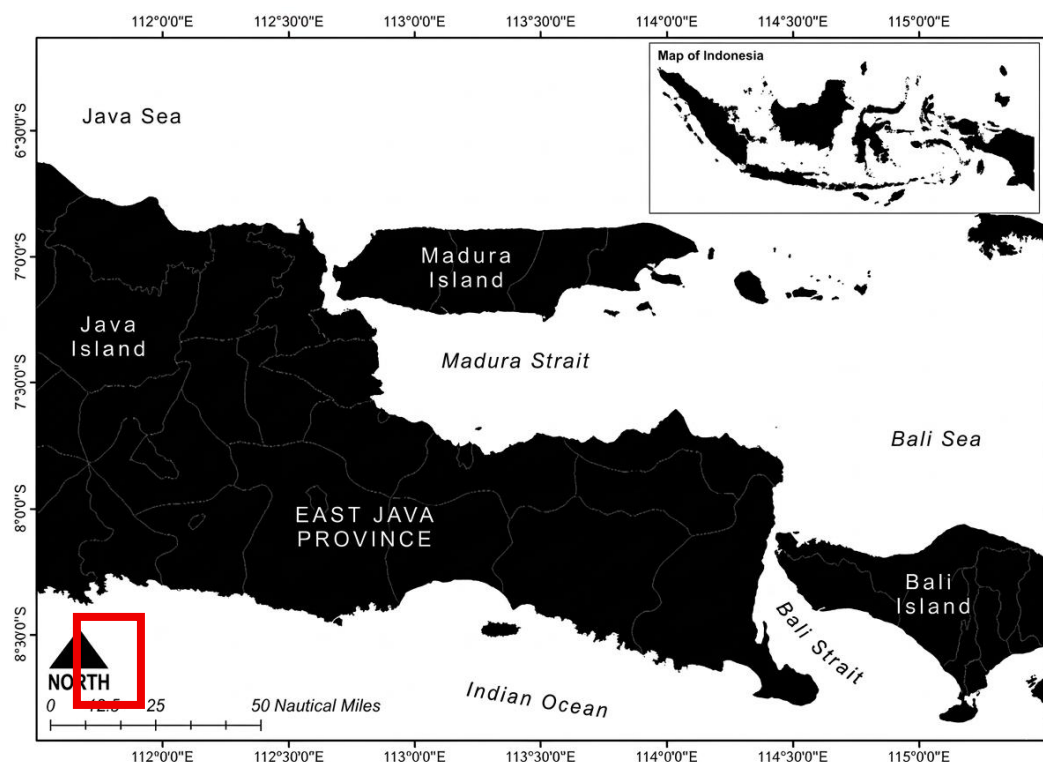


Figure 2. Map of the study area with small fishing ports in East Java

This study integrated FAO-CCRF sustainability indicators with a phase-resolved FI–SI risk-matrix approach. The workflow was structured in sequential steps from the acquisition of field data to parallel sustainability and FI–SI assessments, followed by phase-level overlay, so that each step could build toward a coherent evaluation of ecological performance and operational risk. Primary data came from structured observations and interviews with purse seine crews, with a focus on the highest-exposure phases, i.e., setting, hauling, sorting, and landing. FI (frequency) and SI (severity) scores were assigned using predefined rubric, and scoring was informed by triangulation between observers’ notes, port accident records, and fishers’ self-reports. As regards observers’ involvement and consistency protocols, operational observations were conducted by a team of three trained observers (two primary field observers and one port-based recorder) who completed a joint two-day calibration workshop prior to fieldwork. The calibration involved pilot scoring exercises on archival video footage of purse seine operations, followed by group reconciliation sessions to establish shared interpretation of FI and SI rubric categories. During fieldwork, observers used a standardized scoring codebook with explicit criteria for each frequency and severity level. Daily debriefing sessions were held to resolve the discrepancies in scoring, and consensus scores were recorded when initial observers’ ratings differed by more than one category level. This protocol ensured inter-observer reliability and transparent documentation of scoring decisions. Hazards were identified from direct observation, port accident logs, and fishers’ self-reports, following maritime safety procedures used in recent occupational studies (Türkistanlı & Pehlivan, 2025). Secondary data derived from catch logs, species composition, and landings supplied sustainability indicators: catch per unit effort (CPUE), selectivity, and the share of fish below length at first maturity (L_m), alongside socio-economic contexts covering safety gear access, training, and local governance. This pairing extended decision support beyond standalone OHS matrices

by ensuring that phase-level risk prioritisation was interpreted alongside CCRF-relevant ecological and social trade-offs.

The operationalization of FAO-CCRF indicators in respect of ecological, operational, and social dimensions included selectivity, habitat interaction, safety practices, training, and regulatory compliance. To anchor the selection of indicators in FAO policy language, the indicators were mapped to relevant CCRF provisions (Article 6 covering general principles, Article 7 on fisheries management, and Article 8 on fishing operations). These were benchmarked against CCRF norms and cross-compared with risk levels generated through FI–SI to identify phases or practices that harmed either sustainability performance or human safety (Bolbot et al., 2021). The rationale behind indicator integration was based on the assumption that sustainability and safety were co-produced outcomes of fishing operations rather than independent domains, thus requiring a unified evaluation tool. Importantly, the dual-matrix structure provided a decision-support capability that common fisheries tools often lacked: it linked phase-level sustainability signals to task-level hazard rankings, so as to enable managers to prioritise interventions that delivered dual wins (risk reduction plus improved selectivity/compliance) under constrained resources. In the final analytical stage, a sustainability–risk matrix was developed to consolidate findings and produce actionable recommendations for adaptive management at the port and community level. The integrated matrix aligned mitigation strategies with sustainability principles, thus enabling stakeholders to justify interventions not only on the basis of safety improvements but also through environmental and socio-economic benefits. The feasibility of implementation is addressed in the Discussion by outlining enforcement, labour, and affordability constraints and proposing a tiered sequencing of controls.

3. Results

The surveyed fleet comprised small-scale purse-seine vessels typical of maritime Southeast Asia, with dimensions, materials, and power closely aligned with prior descriptions of handcrafted or semi-industrial units adapted to nearshore pelagic grounds as shown in Table 1. Most boats operated within the 5–30 GT band and used outboard or modest inboard engines suitable for rapid maneuvering around schooling fish in shallow coastal waters, to be consistent with regional practice (Harlyan et al., 2021). Crew complements clustered in the mid-teens and reflected the mixed skill structure documented elsewhere; for instance, experienced skippers and net men paired with family members or local entrants implied shared norms in labor allocation and revenue distribution (Liontakakis et al., 2020). Trips were short, usually day excursions timed to crepuscular activity of pelagic schools and followed by immediate landing to protect quality and cash flow, in order to match operational rhythms reported for artisanal fleets across Indonesia (Agustina et al., 2023; Arnenda & Rochman, 2021; Harlyan et al., 2021). These features were summarised to provide a baseline for interpreting phase-specific risk exposure and CCRF-aligned sustainability performance. To avoid over-generalisation from a cross-sectional dataset, the sea-state context during the sampling window was reported alongside fleet descriptors. Results were interpreted as a snapshot rather than a seasonal average, as the variability of sea state could influence both operational hazards and catch outcomes.

Table 1. Characteristics of small-scale purse seine vessels and operations in East Java

Attribute	Typical Range/Description	Notes for East Java Sample
Vessel size (Gross Tonnage (GT))	5–30 GT	Wooden/Fiber-reinforced plastic hulls; shallow-draft for nearshore grounds
Length Over All (m)	8–15 m	Outboard or small inboard diesel
Crew size (people)	10–20	Family/community crews; task rotation common
Gear	Purse seine (single or two-boat system)	Net depth and bunt reinforced; auxiliary skiff in some sets
Duration of trips	Day trips (≤ 24 h)	Land daily to preserve freshness and price
Main grounds	Nearshore pelagic zones	Bathymetry 10–60 m; seasonal shifts with monsoon
Target species	<i>Sardines</i> , <i>mackerels</i> , <i>small tunas</i>	<i>Auxis rochei</i> , <i>Rastrelliger kanagurta</i> , <i>Scomberoides tol</i> , <i>Katsuwonus pelamis</i> , <i>Euthynnus affinis</i>
Onboard cold chain	Crushed ice in insulated boxes Personal Flotation Devices	“3C+Q” handling encouraged (clean, careful, cool, and quick)
Safety equipment	(PFDs), lights, and Very High Frequency (variable uptake)	Training coverage uneven

Note: Findings represented a sampling-window snapshot across 15-observation days; seasonal variability might influence operational patterns and risk profiles

Operationally, setting, pursing, hauling, and sorting followed the canonical sequence for small purse seines (Figure 3), with higher manual intensity during net deployment and retrieval. Occasional two-boat coordination was observed for scouting and boat-handling support; while this could elevate catch efficiency, it also added communication and collision risks identified in other artisanal contexts (Abreu et al., 2024; Abreu et al., 2022; Saldanha et al., 2020). The vessel profile indicates an operational model that is economically accessible and

socially inclusive for coastal communities, yet remains vulnerable to mechanical strain, environmental variability, and human-centric error. The CCRF-to-phase mapping and FI–SI overlay logic underpinning the subsequent analyses are summarised in Figure 1. Such characteristics emphasize that sustainability challenges in small-scale fisheries extend beyond ecological performance and require consideration of operational safety, design suitability, and community livelihood resilience.

Landings were dominated by small pelagics typical of coastal purse-seine fisheries, with neritic tunas and mackerels comprising over 97% of the catch by weight (Table 2). The assemblage was led by *Auxis rochei* (35.3%), *Rastrelliger kanagurta* (28.3%), *Scomberoides tol* (24%), and *Katsuwonus pelamis* (9%), with a minor share of *Euthynnus affinis* (2.9%). Only a single non-target taxon, *Tylosurus acus*, appeared as bycatch, yielding an overall bycatch rate of ~0.4% by weight. This species composition indicates a high degree of operational selectivity consistent with gear and set strategies designed for schooling pelagics, while still warranting continued attention to mesh configuration and set timing to safeguard size-at-maturity (L_m) compliance across species (Nugraha et al., 2025; Pasa Laksmiana et al., 2024). In CCRF terms, these outcomes map most directly to responsible fishing operations and selectivity objectives (FAO-CCRF Article 8).

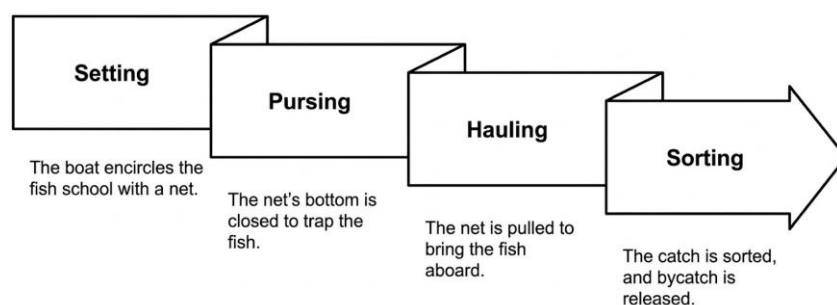


Figure 3. Operational process of small-scale purse seine fishing

Table 2. Species composition, relative abundance (%), and bycatch rate of the purse seine catch

Species (Scientific Name)	Catch (kg)	Relative Abundance (%)	Category
<i>Auxis rochei</i>	6,833	35.3	Main catch
<i>Rastrelliger kanagurta</i>	5,480	28.3	Main catch
<i>Scomberoides tol</i>	4,645	24.0	Main catch
<i>Katsuwonus pelamis</i>	1,735	9.0	Main catch
<i>Euthynnus affinis</i>	570	2.9	Main catch
<i>Tylosurus acus</i>	75	0.4	Bycatch

Where any sub-L_m retention occurs, the ecological signal is a contraction of reproductive potential and a shift in size structure (Table 3), being mechanisms known to depress stock productivity and propagate food-web effects; minimizing such retention remains a priority even under low bycatch conditions (Forestier et al., 2020; Perivolioti et al., 2020; Schijns & Pauly, 2021). In operational terms, persistent removal of undersized fish may contribute to future CPUE declines, which can increase time-at-sea and handling demand, thereby elevating exposure to hazards and serving as a sustainability–safety linkage noted in recent fisheries analyses (Jarernpornnipat et al., 2024). This illustrates the decision-support value of the dual-matrix approach: sub-L_m retention is treated not only as an ecological non-compliance signal but also as a plausible driver of increased safety exposure via greater effort and longer task duration. These findings reinforced the value of pairing FAO-CCRF-aligned selectivity measures with FI–SI risk tracking across phases of setting, pursing, hauling, and sorting to maintain both ecological performance and crew safety. Biological performance, therefore, presented a mixed sustainability signal: while the fishery demonstrated strong environmental selectivity, targeted improvement was still required for long-term stock reproductive viability and responsible catch profiling.

A CCRF-aligned indicator matrix in Table 4 was assembled to benchmark ecological status, bycatch management, social safeguards, and governance compliance (Atufa et al., 2023). To strengthen transparency and reproducibility, each indicator was explicitly linked to relevant FAO-CCRF provisions (Articles 6–8) and mapped to the operational phase(s) where it manifested most strongly. Ecologically, fleets with higher L_m adherence and lower bycatch ratios scored favorably, while operations with frequent juvenile retention or weak discard protocols scored lower. On the social and governance axes, availability of Personal Protective Equipment, basic safety training, and adherence to licensing/zoning strengthened scores; conversely, gaps in enforcement and emergency preparedness depressed them, thus reflecting constraints reported for small-scale fisheries (Malik et al., 2019). The gaps of these indicators are carried forward to the discussion of implementation, in order to recognize the proposition that ecological inefficiencies could indirectly escalate occupational risk, rendering sustainability not

merely a conservation target but a contributor to safe operational workflow.

Table 3. Summary of catch composition and selectivity performance of dominant pelagic species

Species (Scientific Name)	Lm (cm)	Immature (%)	Mature (%)	Basis in Manuscript
<i>Rastrelliger kanagurta</i>	21.0	0	100	From length classes; all $\geq$ Lm
<i>Auxis rochei</i>	25.8	34	66	From length-frequency counts vs. Lm
<i>Euthynnus affinis</i>	42.3	100	0	All observed $<$ Lm
<i>Scomberoides tol</i>	26.0	2	98	From length-frequency counts vs. Lm
<i>Katsuwonus pelamis</i>	41.3	0	100	All observed $\geq$ Lm
<i>Tylosurus acus (bycatch)</i>	47.3	0	100	All observed $\geq$ Lm

Note: Data reflected the sampling window; Lm compliance patterns might vary with seasonal fishing grounds and gear practices

Table 4. Food and Agriculture Organization Code of Conduct for Responsible Fisheries (FAO-CCRF) sustainability indicator matrix for small-scale purse seine operations

Dimension	Indicator	Metric/Evidence	Compliance Status	Action Cue
Ecological	Lm compliance	Share of catch $\geq$ Lm	Mixed by species	Reinforce mesh/selective tactics where sub-Lm is present
Ecological	Bycatch ratio	% of non-target/bycatch	Low to moderate	Continue avoidance protocols; monitor
Ecological	Habitat interaction	Seabed contact risk	Low (pelagic gear)	Maintain depth/bathymetry safeguards
Operational	Trip efficiency	CPUE, haul success	—	Optimize set timing; two-boat coordination
Social	Safety capacity	Training, PFD usage	Uneven	Targeted OHS training; audits
Governance	Adherence to rules	Zoning/licensing, mesh regs	Generally compliant	Periodic inspections; fisher dialogue

Risk computation followed a $FI \times SI$ scheme to quantify operational hazards. The calculation framework and scaling boundaries were provided for transparency and reproducibility. FI and SI scores were assigned using a predefined rubric and triangulated evidence (observer logs, port records, and fishers' reports); when multiple scorers were involved, consistency was supported through calibration and a shared codebook. Tables 5–7 define the FI/SI scales and the resulting risk-index bands used in subsequent matrices. Hazards were positioned in a 2D matrix to classify low/medium/high risk bands and to enable rapid prioritization (Bolbot et al., 2021; Ivančan & Lisjak, 2021). The reference structure used for categorization is shown for illustration.

Table 5. Calculation framework of Frequency Index (FI) and Severity Index (SI)

Component	Scale	Description
FI	1–5	Rare (1), Occasional (2), Frequent (3), Very frequent (4), Persistent (5)
SI	1–5	Insignificant (1), Minor injury/delay (2), Serious injury/Major damage (3), Permanent disability/Large loss (4), Fatality/Catastrophic loss (5)
Risk Index (RI)	$FI \times SI$	1–4 Low; 5–9 Moderate; 10–16 High; 17–25 Extreme
Use	—	Map hazards to RI; prioritize controls proportionally

Table 6. Accident severity scale (SI)

SI	Description of Consequences	Typical Examples
1	No injury; negligible loss	Brief delay; superficial abrasion
2	Minor injury; first-aid only; small repair	Rope burn; small cut; minor gear tear
3	Serious injury; medical treatment; major repair	Fracture; deep laceration; winch failure
4	Permanent disability; very large loss	Amputation; total net loss; engine wreck
5	Fatality or multiple casualties; catastrophic loss	Man-overboard fatality; vessel loss

Table 7. Accident frequency categories (FI)

FI	Likelihood Descriptor	Qualitative Frequency
1	Rare	$\leq$ once per several years
2	Occasional	A few times per year
3	Frequent	Monthly to bi-weekly
4	Very frequent	Weekly
5	Persistent	Daily or per trip

The integration phase of the framework culminated in the development of Operational Risk and Safety. Table 8 consolidates the FI–SI outputs by hazard and phase and links each control option to an implication of CCRF-relevant sustainability. FI–SI scoring highlighted distinct risk peaks around rope handling during set and purse closure, and around hauling on wet and cluttered decks. Slips/trips under spray, sudden vessel dynamics, and rope/lead-line entanglements contributed the largest share of high-risk cells, to be consistent with accident typologies described for small fishing ports and artisanal fleets (Cengiz, 2022; Irvana et al., 2020; Özaydın et al., 2022). Weather-related spikes in wave and wind amplified risks during night hauling and cross-deck transfers, in order to echo findings that adverse sea states elevated capsizing/drowning probabilities and required active avoidance or tight operational control (Alwi et al., 2020; Lee et al., 2024; Watson et al., 2022).

Task-level mitigations emphasized communication discipline in two-boat maneuvers, spacing and sequencing during pursing, anti-slip deck treatments, rope guides/guards, and fatigue management through micro-breaks and rotation, i.e., interventions shown to reduce the likelihood of errors and severity of injuries in comparable artisanal contexts (Saldanha et al., 2020; Türkistanlı & Pehlivan, 2025). To clarify novelty relative to existing tools, the framework was neither a standalone OHS matrix nor a certification-style indicator set; it was a combined decision-support output that (i) identified phase-level risk peaks (FI–SI) and (ii) simultaneously flagged where CCRF-aligned sustainability indicators were compromised, to enable prioritisation of interventions with dual ecological and safety benefits under limited resources. Unlike assessment models that treated sustainability and safety as separate domains, this study positioned them as interdependent components of long-term fishery viability.

Table 8. Operational risk and safety management matrix for small-scale purse seine operations in East Java

Hazard/Phase	FI	SI	RI (Class)	Key Controls	Sustainability Linkage
Rope handling during setting/pursing	3	3	9 (Moderate)	Gloves, fairleads, line bins, crew signals	Fewer injuries → less downtime/waste
Net deployment in rough sea	2–3	4	8–12 (High)	Weather gate, go/no-go check, two-boat comms	Avoid gear loss/marine debris
Hauling (mechanical strain)	3–4	3	9–12 (High)	Hauler guards, pace/rotation, task zoning	Reduce mishandling/bycatch stress
Sorting on wet deck	4	2	8 (Moderate)	Anti-slip, headlamps, bin ergonomics	Faster, cleaner handling preserves quality
Fatigue (back-to-back trips)	3	3	9 (Moderate)	Rest rules, watchbill, caffeine limits	Maintain decision quality/CPUE
Night navigation/traffic	2	5	10 (High)	Automatic Identification System/lights, lookout, approach standard operating procedures	Prevent collision-linked spills/loss
Ice handling and cold-chain	2	2	4 (Low)	Box insulation, early icing, temp checks	Higher ex-vessel value; food safety
Sub-Lm retention risk	NA	NA		Mesh/escape panels; release protocol	Protect spawning biomass; long-run CPUE

To connect ecological compliance with safety exposure, FI–SI risk levels were overlaid with CCRF indicators at the level of fishing phases (setting, pursing, hauling, and sorting). The integrated view (Table 9) revealed two robust patterns. First, phases with better selectivity and Lm adherence tended to show lower cumulative time on task and fewer handling complications, thereby reducing FI scores. This supports evidence that selective practices and gear innovations could simultaneously improve the ecological and human-factor outcomes by reducing unnecessary catch processing and on-deck congestion. Second, phases associated with higher bycatch and sub-Lm capture correlated with longer sorting times, more manual re-handling, and elevated entanglement/knife-handling risks, indicating that ecological noncompliance could propagate into higher safety exposure through the complexity of work process. This phase-resolved overlay was the central decision-support contribution relative to certification-style indicator sets (e.g., MSC-type schemes), EAFM scorecards, or standalone OHS matrices, which often did not translate performance signals into task-specific prioritisation of hazards. The resulting management cues emphasise low-cost procedural controls first (role clarity, sequencing, and deck order), followed by incremental retrofits where feasible, in order to reduce barriers to adoption in data-limited and resource-constrained fishing communities.

Tables 8 and 9 consolidate the FI–SI results and corresponding sustainability indicators into a dual-matrix decision-support output. These matrices synthesized operational hazards, frequency–severity classifications, and mitigation priorities while explicitly linking each intervention to FAO-CCRF sustainability domains. For instance, hauling and sorting phases, identified as high-risk based on elevated $FI \times SI$ scores, were also associated with ecological inefficiencies such as increased bycatch handling burden and reduced adherence to Lm thresholds. The key novelty was operational translation: sustainability signals (Lm noncompliance, handling burden, and compliance gaps) were mapped into phase-specific safety consequences (time-at-risk, re-handling, and knife/rope exposure) and therefore prioritised controls. The dual-table system demonstrated that technical interventions (e.g.,

ergonomic net-hauling aids and improved deck layout) and behavioral measures (e.g., safety training and crew rotation) could reduce both occupational risk and ecological stress. As summarized in Tables 8 and 9, this integrated approach allows fishery managers and local cooperatives to visualize the coupling between safety and sustainability outcomes, prioritize mitigation strategies, and ensure alignment with FAO-CCRF principles and adaptive maritime safety frameworks (Atufa et al., 2023; Mohsin et al., 2020). Implementation constraints (enforcement capacity, informal labour arrangements, and affordability) are addressed explicitly in the Discussion, in order to justify a tiered sequencing of controls rather than assuming frictionless uptake.

Table 9. Integration of sustainability compliance and operational risk levels by fishing phase

Phase	Typical FI–SI Profile	CCRF Compliance Signals	Joint Interpretation	Management Cue
Setting	FI 2–3/SI 3 (Moderate)	Low habitat impact; bycatch depends on school ID	Risk rises with sea state and haste	Pre-set checks; school verification
Pursing	FI 3/SI 3–4 (High)	Selectivity driven by mesh/bunt	Communication critical; entanglement risk	Clear roles; safe radio/hand signals
Hauling	FI 3–4/SI 3 (High)	Handling stress may raise bycatch	High mechanical strain; slips	Guarding, rotation, pace control
Sorting	FI 4/SI 2 (Moderate)	Lm compliance depends on practice	Wet-deck slips; quality risks	Anti-slip; sub-Lm release protocol
Landing	FI 2/SI 2 (Low)	Continuity of the cold chain	Pinch/crush hazards minimal	Ramp SOP; rapid icing transfer

Three sensitivities emerged with management relevance. First, crew size and coordination: mid-teen crews supported efficient purse-closure and heavy lifts, but any erosion in role clarity increased collision and line-handling errors, thus reinforcing the need for standardized task briefings and hand-signal/voice protocols, especially in two-boat operations where miscommunication was a documented precursor to incidents (Abreu et al., 2024; Saldanha et al., 2020). Second, trip duration and timing: short and daily trips reduced the accumulation of fatigue and weather exposure, thus echoing safety advantages reported in artisanal contexts (Ruenes et al., 2024). However, repeated back-to-back sorties without rest might still aggregate the risk of fatigue; schedule management therefore remained essential. Because the dataset was cross-sectional, results reflected the sampling window; seasonal sea-state conditions were reported and interpretation avoided treating observed FI and ecological patterns as seasonal averages. Seasonal replication of the FI–SI scoring (monsoon vs. calmer periods) was identified as a priority for future work. Third, mesh size and gear configuration: fleets using meshes and purse-line that promoted Lm compliance displayed lower sorting burden and fewer handling incidents, to be consistent with the literature linking mesh optimization to reductions in juvenile capture and bycatch complexity (Pasa Laksmiana et al., 2024). In areas with complex bathymetry or near sensitive habitats, shallow-water sets increased snagging potential, thus underlining the value of depth awareness and area-based controls to minimize gear–seabed interaction (Sipahutar et al., 2022).

Taken together, the results substantiated two integrative claims. First, ecological selectivity and safety were jointly attainable: adopting Lm-compliant meshes, BRDs, and phase-specific handling protocols improved stock stewardship while reducing FI–SI risk loads by simplifying deck work and shortening hazardous phases (Pons et al., 2022). Second, CCRF-aligned governance levers were consistent with safety payoffs: licensing and zoning compliance, PPE availability, and basic safety training core elements in CCRF practice aligned with profiles of lower risk and better environmental indicators, thus supporting the practical utility of CCRF metrics as dual sustainability–safety signals (Atufa et al., 2023; García & Duarte, 2023).

4. Discussion

The operational profile, including short trips, nearshore sets, and mixed-skill crews working on compact and handcrafted vessels, was documented for small-scale purse seiners in East Java. This helped explain the hazard patterns revealed by the FI–SI analysis. Similar artisanal configurations across maritime Southeast Asia rely on intense manual handling during setting, pursing, and hauling, which elevates the likelihood of rope burns, entanglement, and slips on wet decks, especially under time pressure and variable sea state. This pattern is consistent with accident typologies described for small fishing ports and artisanal fleets (Cengiz, 2022; Irvana et al., 2020; Özaydın et al., 2022). The prevalence of day trips likely limits cumulative fatigue, but rapid turnarounds could compress recovery and concentrate hazardous tasks into narrow operational windows. This dynamic echoes evidence that short sorties reduced weather exposure yet could still aggregate risk without rest discipline (Alwi et al., 2020; Lee et al., 2024; Watson et al., 2022). Two-boat coordination improved search efficiency but introduced communication-sensitive failure modes such as near-miss collisions, gear fouling, and mis-sequenced purse closure. These observations align with findings that multi-vessel maneuvers demand explicit protocols to avoid

escalation from frequent and low-severity incidents to rarer but more severe events (Abreu et al., 2024; Saldanha et al., 2020). Importantly, exposure to hazards and ecological performance interacted: operational choices that increased re-handling (e.g., sub-Lm capture requiring sorting/release) also increased time-at-risk and could raise FI, in order to reinforce the co-production logic that the dual-matrix framework rendered operationally visible.

The length–frequency results showed that when mesh size and gear configuration aligned with species’ length at first maturity (Lm), catches skewed toward mature fish and sorting was therefore faster, cleaner, and less injurious. This pattern mirrors experimental and regulatory literature which demonstrated that mesh optimization reduced juvenile capture and stabilized stock productivity (Nugraha et al., 2025). Such optimization also simplified on-deck handling and reduced operational complexity (Pasa Laksmiana et al., 2024; Sipahutar et al., 2022). Conversely, persistent retention below Lm reduced reproductive output and shifted population structure, which could depress future catch rates. These ecological consequences were documented across multiple fishery systems (Forestier et al., 2020; Moore et al., 2021; Perivolioti et al., 2020; Schijns & Pauly, 2021). Operationally, meeting the same volume targets then required longer search and handling time, thus increasing exposure to entanglement, cuts, and weather events. The results align with the notion that ecological noncompliance could propagate into higher FI scores via extended hazardous phases (Jarernpornnipat et al., 2024). This coupling reinforced a practical message for port managers: selectivity was not only a conservation lever but also a frontline safety intervention because it reduced unnecessary re-handling and decongested decks. In CCRF terms, these mechanisms clarified why responsible fishing operations indicators (FAO-CCRF Article 8) could function as dual ecological–safety signals when evaluated at the operational-phase level.

Selective technology further strengthened this nexus. Bycatch Reduction Devices and size-selective practices reduced unintended catch and shortened sorting time, to lower near-term potential of incidents while buffering stocks (Pons et al., 2022). The empirical pattern that phased with better Lm adherence had lower composite risk; this suggested that training and enforcement around gear configuration and size limits were as relevant to occupational safety as they were to stock rebuilding. The finding that risk peaks clustered in predictable tasks and conditions supports an adaptive management cycle which focuses on phase-specific learning. In this study, phase-level FI–SI scoring was assigned using a standardized rubric and triangulated through structured interviews and port documentation; scoring consistency was maintained through shared coding rules and debrief-based reconciliation wherever more than one observer was involved. Routine re-scoring by phase, interpreted with crews, could support “plan–do–check–adjust” learning by identifying high-yield mitigations. Examples included deck anti-slip treatments in hauling lanes, rope guides for lead-lines, and standardized hand signals for two-boat sets. Such iterative assessment allowed managers to evaluate whether risk levels declined after interventions were implemented. This approach aligns with the emphasis of adaptive management on feedback loops that integrated local ecological knowledge with scientific metrics to refine practice under environmental variability (Galappaththi et al., 2022; Goldemberg, 2015; Karr et al., 2021; Mulyasari et al., 2025). Embedding crews in joint diagnosis and solution design was also a safety amplifier. Participation improved hazard recognition, strengthened protocol ownership, and raised compliance, resulting in outcomes that had been linked to lower accident rates in small-scale fisheries (Hines et al., 2020; Rehren et al., 2021). Co-management arrangements and local committees provided practical venues to institutionalize this cycle and to balance ecological and occupational priorities at the port level (Farella et al., 2021; Kapembwa et al., 2021; Oloruntuyi et al., 2023).

Linking FI–SI outputs to FAO-CCRF sustainability domains revealed actionable synergies. The operational matrix (Table 8) aligns concrete mitigations with CCRF ecological, technical, and social-governance pillars. Specific measures include ergonomic net-hauling aids, improved deck drainage and layout, crew rotation, and focused toolbox talks, which address bycatch limits, Lm adherence, gear optimization, safety training, and compliance (Atufa et al., 2023; García & Duarte, 2023). To anchor this in FAO policy language, the mapped domains are referenced against specific CCRF Articles (Articles 6–8) and their indicators. The complementary integration table (Table 9) makes the coupling explicit. Interventions that reduce entanglement risk and time-at-risk also tend to reduce bycatch handling burden and improve Lm compliance, in order to yield concurrent gains in safety and sustainability (Mohsin et al., 2020). In practical terms, prioritizing mitigations in high-risk/high-impact cells offers a transparent and phase-specific roadmap for cooperatives and port managers. The feasibility of implementation, however, is shaped by local constraints (enforcement capacity, informal labour arrangements, and affordability). These constraints justify a practical sequencing of interventions, whereby low-cost procedural measures (e.g., toolbox talks, role clarification, and task briefings) and targeted training are implemented first, followed by incremental retrofits as resources allow.

From a maritime-technology perspective, modest retrofits such as anti-slip surfacing, guarded fairleads, color-coded lines, and low-inertia net capstans were well matched to small-port realities and directly targeted the high FI×SI phases identified in this study. Coupled with cold-chain discipline and organized holds, these changes also reduced on-deck clutter and handling time, thus indirectly shrinking exposure windows while protecting product value. Accordingly, mitigation options were organized as a tiered package (procedural controls and training → low-cost retrofits → higher-cost mechanisation) to support adoption even where affordability and enforcement were limited. At the port level, embedding the dual matrix into routine safety meetings provided a shared evidence

base for prioritising investments and tracking progress against CCRF-consistent indicators.

Overall, the study demonstrated that safety and sustainability were not competing objectives but co-produced outcomes of the same operational decisions. Concentrating mitigations where FI×SI was highest and *Lm* adherence/bycatch performance was weakest delivered outsized returns. The paradox of selective fishing cautions that gear-only solutions could displace rather than solve problems; therefore, integrated and phase-specific measures that blended technical, behavioral, and governance levers were required. To avoid over-generalisation, the framework was presented as transferable in structure but not in score values: application beyond East Java or purse seine operations required re-scoring and local validation. By unifying FI–SI analytics with FAO-CCRF indicators, the framework offered a practical pathway for small ports in East Java to improve crew safety, stabilize ecological performance, and strengthen fishery resilience under environmental variability.

5. Conclusions

This study showed that integrating FAO-CCRF indicators with an FI–SI operational risk framework provided a phase-based decision-support tool for small-scale purse seine fisheries. Higher selectivity and *Lm* compliance were associated with shorter handling time and lower safety exposure, whereas sub-*Lm* retention and prolonged sorting increased re-handling, entanglement risk, and fatigue, thus supporting the conclusion that sustainability and safety were co-produced in fishing operations. The dual-matrix output supports low-cost and adoptable interventions (procedures, training, and targeted equipment improvements) that could improve ecological performance while reducing occupational hazards. Limitations: findings were specific to East Java and purse seine operations, and FI–SI estimates were semi-quantitative and derived from a cross-sectional snapshot; thus, the framework was transferable in structure, but score values required re-scoring and local validation in other settings. Future work should test the approach across seasons (monsoon vs. calmer periods), extend it to other gears, and evaluate cost–benefit and participatory monitoring to strengthen generalisability and implementation.

Author Contributions

Conceptualization and methodology, S. and E.S.Y.; investigation and data collection, C.S.U.D.; formal analysis, S. and M.K.R.; writing—original draft preparation, S.; writing—review and editing, M.K., E.S.Y., and F.; supervision and funding acquisition, F. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data supporting the findings of this study are not publicly available because they contain confidential field and operational information related to fishers and port activities. Therefore, the dataset is not shared to protect participant confidentiality and site-specific operational sensitivity.

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Conflicts of Interest

The authors declare no conflict of interest.

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